

# *Bletilla striata* Polysaccharide-/Chitosan-Based Self-Healing Hydrogel with Enhanced Photothermal Effect for Rapid Healing of Diabetic Infected Wounds via the Regulation of Microenvironment

Kai Zhao, Zhengbo Hu, Xingcan Chen, Yuchi Chen, Mingyuan Zhou, Xiaoqing Ye, Fangmei Zhou, Bingqi Zhu,\* and Zhishan Ding\*



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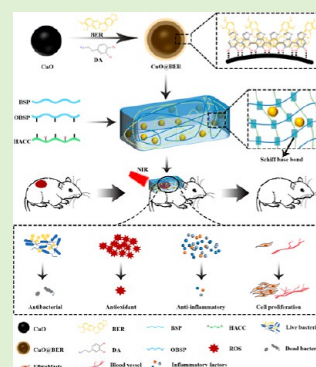
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**ABSTRACT:** The management of diabetic ulcers poses a significant challenge worldwide, and persistent hyperglycemia makes patients susceptible to bacterial infections. Unfortunately, the overuse of antibiotics may lead to drug resistance and prolonged infections, contributing to chronic inflammation and hindering the healing process. To address these issues, a photothermal therapy technique was incorporated in the preparation of wound dressings. This innovative solution involved the formulation of a self-healing and injectable hydrogel matrix based on the Schiff base structure formed between the oxidized *Bletilla striata* polysaccharide (BSP) and hydroxypropyltrimethylammonium chloride chitosan. Furthermore, the introduction of CuO nanoparticles encapsulated in polydopamine imparted excellent photothermal properties to the hydrogel, which promoted the release of berberine (BER) loaded on the nanoparticles and boosted the antibacterial performance. In addition to providing a reliable physical protection to the wound, the developed hydrogel, which integrated the herbal components of BSP and BER, effectively accelerated wound closure via microenvironment regulation, including alleviated inflammatory reaction, stimulated re-epithelialization, and reduced oxidative stress based on the promising results from cell and animal experiments. These impressive outcomes highlighted their clinical potential in safeguarding the wound against bacterial intrusion and managing diabetic ulcers.



## 1. INTRODUCTION

Diabetic ulcer (DU) is one of the most serious complications of diabetes and is the primary cause of lower limb disability among diabetic patients.<sup>1</sup> The wound environment of DU is complex, often featuring bacterial infections, chronic inflammation, and elevated levels of oxidative stress.<sup>2</sup> The combination of the above factors contributes to the persistence of DU, causing challenges to the clinical treatment. Conventional dressings like gauze were inadequate in solving such complex DU wounds.<sup>3</sup> Consequently, to effectively facilitate the healing of DU, the key lies in designing multifunctional biomaterials tailored to the intricate microenvironment of DU wounds.

Persistent hyperglycemia makes DU wounds highly susceptible to bacterial growth. Although antibiotics are commonly used to solve this issue, their prolonged usage can lead to the emergence of drug-resistant bacteria, significantly compromising their therapeutic effectiveness.<sup>4</sup> In recent years, researchers have turned their attention to photothermal therapy (PTT), which directly destroy bacterial structures rather than interfering metabolism, as a promising alternative to antibiotics.<sup>5,6</sup> Photothermal agent (PTA) that converts light into heat is the key to the implementation of PTT.<sup>7</sup> Many inorganic PTAs have excellent photothermal effect and stability; however, the potential biotoxicity restricts their application.<sup>8</sup> On the other hand, some organic PTAs exhibit

good biocompatibility but unsatisfied photostability.<sup>9</sup> To achieve a more comprehensive PTA, the design of inorganic–organic PTAs has emerged, aiming to integrate exceptional photothermal effects into high biocompatibility. Copper oxide nanoparticles (CuO NPs) are particularly interesting due to their positive effects on anti-inflammatory, angiogenesis and wound healing, in part because copper is an important trace element.<sup>10,11</sup> Polydopamine (PDA) is one of the most promising organic PTAs because of its excellent photothermal conversion efficiency and biocompatibility.<sup>12</sup> Moreover, PDA possesses multiple functionalities, including drug-carrying and adhesion capabilities.<sup>9</sup> Therefore, dopamine-coated CuO NPs provide a feasible approach for the development of a biosafe and effective PTA.

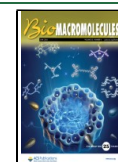
Persistent infection and excessive reactive oxygen species (ROS) resulting from hyperglycemia can exacerbate inflammatory reactions, leading to oxidative damage and impaired immune cell function, ultimately hindering wound healing.<sup>2,13</sup>

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Berberine (BER) is an isoquinoline alkaloid and exhibits a range of pharmacological activities, including antibacterial, anti-inflammatory, antioxidant, and regulation of glucose and lipid metabolism.<sup>14</sup> These properties make it a promising candidate for alleviating inflammation and oxidative stress in DU. Interestingly, it has been reported that the combination of metal NPs with BER achieved enhanced photothermal performance, resulting in the improved antibacterial and antitumor efficacy.<sup>15,16</sup> Therefore, the combination of BER with dopamine-coated CuO NPs holds great promise for achieving enhanced PTA outcomes.

Self-healing hydrogels possess remarkable properties as they can maintain their structural integrity when subjected to external tension and mechanical stress, making them superior dressing for tissue repair and regeneration.<sup>17,18</sup> The Schiff base structure has been widely used in the development of self-healing hydrogels. Polysaccharide is an ideal compound for preparing Schiff base hydrogels, either directly or after modification.<sup>19–21</sup> In addition to the good biocompatibility, nontoxicity, and antibacterial properties of chitosan, hydroxypropyltrimethylammonium chloride chitosan (HACC) exhibits not only improved water solubility and adhesive property but also enhanced antioxidative and antibacterial characteristics.<sup>22</sup> *Bletilla striata* polysaccharide (BSP) is the main active ingredient found in *B. striata* (Thumb.) Reichb.f. which has pharmacological activities of astringent, detumescence, and promoting tissue generation.<sup>23</sup>

Inspired from the above, a novel approach was adopted where CuO NPs were coated with dopamine and BER to create a novel composite PTA (CuO@BER). Subsequently, this PTA was incorporated into a dynamic cross-linked hydrogel formed by OBSP and HACC, resulting in the development of a multifunctional composite hydrogel (CuO@BER/BH). The research included studies of various aspects. First, the physical and chemical properties of the hydrogel were studied in detail. In addition, the photothermal properties, antibacterial activity, and in vitro anti-inflammatory and antioxidant properties of hydrogels were fully investigated. Lastly, the wound healing efficacy of the hydrogel was evaluated by using a diabetic mouse model with infected wounds to evaluate the effectiveness of the hydrogel in wound healing. The comprehensive assessments offered scientific data on the performance of the hydrogel in promoting wound healing under an intricate microenvironment of DU wounds.

## 2. MATERIALS AND METHODS

**2.1. Materials.** BSP, HACC, hydroxylamine hydrochloride, dopamine hydrochloride, BER hydrochloride, and KBr were obtained from Yuanye Biotechnology Company (Shanghai, China). D<sub>2</sub>O and CuO NPs were purchased from Macklin Biochemical Co., Ltd. (Shanghai, China). NaIO<sub>4</sub>, sheep red blood cells, and CCK-8 were obtained from Shanghai Huake Labware Co., Ltd. (Shanghai, China), Nanjing SenBeijia Biological Technology Co., Ltd. (Nanjing, China), and biosharp (Beijing, China), respectively. Dulbecco's modified Eagle's medium (DMEM), fetal bovine serum, penicillin, and streptomycin were purchased from Gibco (Waltham, MA, USA). Flow cytometry kit was purchased from Becton Dickinson Company (Waltham, MA, USA), and streptozotocin (STZ) and the ROS detection kit were obtained from Beyotime Biotechnology Co., Ltd. (Shanghai, China). Bacterial vitality/toxicity testing kit and Calcein-AM/PI double stain kit were purchased from Solarbio Science & Technology (Beijing, China) and Yeasen Biotechnology (Xiamen, China).

**2.2. Synthesis and Characterization of Oxidized *B. striata* Polysaccharides (OBSP).** BSP was fully dissolved in deionized water

to obtain a 1% solution. Subsequently, a certain amount of NaIO<sub>4</sub> was added at a molar ratio of 1:0.4 to BSP before the reaction in the dark for 24 h. Equimolar amounts of ethylene glycol were then slowly added, and the solution was stirred continuously for 2 h to quench the reaction. The solution was centrifuged at 4000 rpm for 10 min, and the precipitate was discarded to retain the supernatant, which was then dialyzed using a dialysis bag with a molecular weight cutoff of 3000 Da for 72 h. Finally, after dialysis, the supernatant was collected and freeze-dried for 48 h to obtain OBSP. Hydroxylammonium chloride titration method was used to measure the oxidation degree of OBSP.<sup>24</sup> At the same time, BSP and OBSP were analyzed by Fourier transform infrared spectroscopy (FTIR, Nicolet iS50, Thermo Fisher Scientific, USA) using the KBr compression method. After three dissolution freeze-drying exchanges using D<sub>2</sub>O, <sup>1</sup>H NMR detection was performed on a Bruker AVANCE III 600 MHz Digital NMR spectrometer (Bruker, Germany).

**2.3. Synthesis and Characterization of CuO@BER NPs.** After fully dissolving 100 mg of dopamine and 50 mg of BER in 100 mL of deionized water, 500 mg of CuO NPs was dispersed in the above solution and stirred for 24 h. CuO@BER NPs were collected after centrifugation and freeze-drying. Scanning electron microscopy (SEM, SU8010, Hitachi, Japan), transmission electron microscopy (TEM, H-7650, Hitachi, Japan), and X-ray energy-dispersive spectroscopy (EDS) were used to characterize the morphology, structure, and elemental composition of the nanoparticles.

**2.4. Photothermal Properties.** For the photothermal properties of CuO NPs and CuO@BER NPs, 200  $\mu$ L of CuO NPs and CuO@BER NP dispersion solution (1 mg/mL) were transferred into a centrifuge tube and irradiated by NIR (1 W/cm<sup>2</sup> 808 nm) laser for 10 min. The temperature was recorded every minute, and deionized water was used as the control. At the same time, the photothermal conversion efficiency ( $\eta$ ) of NPs was calculated according to the method reported previously.<sup>25,26</sup> Subsequently, the photothermal performance of CuO NPs and CuO@BER NP dispersions at different concentrations was also evaluated.

For the photothermal properties of hydrogels, 0.5 mL hydrogels were placed in 1.5 mL centrifuge tubes and centrifuged to the bottom of the tubes. The NIR laser under the same parameters as above and thermal camera were used for irradiation and recording. BH hydrogel and CuO/BH hydrogel were used as the control, and CuO/OBSP/HACC composite hydrogel (CuO/BH) was obtained by dispersing CuO NPs into the OBSP solution and then mixing and stirring with the HACC solution (the final concentration of CuO NPs was 1 mg/mL). In order to observe the photothermal stability of the hydrogel, the NIR irradiation cycle with 5 min on and 5 min off was repeated four times, and the temperature was recorded every minute.

**2.5. Synthesis and Characterization of Composite Hydrogels.** OBSP and HACC were separately dissolved in deionized water to prepare 3% solutions, and then they were mixed at equal volumes and stirred evenly until the OBSP/HACC composite hydrogel (BH) was obtained. CuO@BER/OBSP/HACC composite hydrogels (CuO@BER/BH) were obtained by dispersing CuO@BER NPs into the OBSP solution and then mixing and stirring with HACC solution (the final concentration of CuO@BER NPs was 1 mg/mL).

The obtained hydrogel was vacuum-freeze-dried to obtain the dehydrated sample. The cross section of the sample was taken and sprayed with gold for microscopic morphology observation by SEM. After grinding, the sample powder was blended with KBr to prepare compressed pellets for FTIR measurements at 4000–500 cm<sup>-1</sup>.

The rheology characteristics of the hydrogel were measured by a rotational rheometer (MCR 302, Anton Paar, Austria). The camera was used to record the macro injectability and self-healing of hydrogels. The shape adaptability of the hydrogel was evaluated using molds of different shapes. Different biological and nonbiological objects were used to observe the adhesion of hydrogels. The water content and swelling rate of the hydrogel were determined by the methods according to the previous reports.<sup>27,28</sup>

**2.6. Antibacterial Activity.** *Staphylococcus aureus* (*S. aureus*, ATCC25923), *Escherichia coli* (*Escherichia coli*, ATCC25922), and *Pseudomonas aeruginosa* (*P. aeruginosa*, ATCC27853) were used in

our study. After the enrichment culture of single colony, the bacterial solution was diluted to  $10^6$  CFU/mL. The hydrogel was irradiated by ultraviolet (UV) light for 2 h in advance. 100  $\mu$ L of hydrogel was added to the 24-well plate, and then 100  $\mu$ L of bacterial liquid was added. 100  $\mu$ L of sterile physiological saline was used as the control group. Subsequently, the NIR group and non-NIR group were subjected to 5 min of NIR treatment or nontreatment, respectively. The plate was then placed in an incubator for further cultivation of 3 h, and the surviving bacteria were successively resuspended with 900  $\mu$ L of sterile physiological saline, diluted, and dropped onto the solid culture medium for 18 h cultivation.

The turbidity method of bacterial solution was used for the quantitative detection of the bacterial survival rate. The bacteria solution was treated with hydrogel and NIR radiation using the same method mentioned above. After incubation for 3 h, 900  $\mu$ L of sterile physiological saline was added to suspend the bacterial solution, and then 100  $\mu$ L of the suspension was transferred to a test tube containing 3 mL of culture medium. After the suspension was incubated for another 18 h, the absorbance of the suspension was measured at 600 nm.

The bacteria solution treated by NIR irradiation was collected, fixed by glutaraldehyde, and dehydrated by graded ethanol solution. Subsequently, the bacterial solutions were dropped onto the silicon wafer and dehydrated, and SEM was used to observe the morphology of the bacteria.

The antibiofilm activity of the hydrogel was evaluated by crystal violet staining according to the method previously reported.<sup>29</sup> The procedures of bacterial culture, washing, bacterial concentration adjustment, and cocultivation were carried out in the same manner as those of the plate method. Subsequently, 900  $\mu$ L of sterile physiological saline was added to each well for the resuspension of bacteria. 100  $\mu$ L of the resuspended liquid was transferred to a new 24-well plate, followed by the addition of 1 mL of fresh culture medium to each well. The cultures were then incubated for 72 h, with the medium changed every 24 h. After the 72 h cultivation period, the biofilm was gently washed three times with sterile physiological saline. Following this, 200  $\mu$ L of paraformaldehyde was added for a fixation of 15 min, after which the biofilm was washed again three times with sterile physiological saline. Next, 300  $\mu$ L of crystal violet dye was added to each well, and the plate was incubated for 15 min. Upon completion of the staining, the excess dye was washed away with sterile physiological saline. Finally, 200  $\mu$ L of a 33% acetic acid solution was added to dissolve the crystal violet dye, and the absorbance of the resulting solution was measured at 570 nm by using a microplate reader to quantify the biofilm amount.

**2.7. In Vitro DPPH Clearance Assay.** The antioxidant capacity of the hydrogel was determined by the DPPH radical scavenging method. After mixing 2 mL of hydrogel PBS solution and 2 mL of DPPH solution at 25 °C, they were placed in a dark environment for 30 min. Subsequently, the absorbance of the sample was measured at 517 nm by using a microplate reader (BioTek Epoch 2 USA). 1 mg/mL portion of vitamin C solution was used as a positive control. The percentage of DPPH free-radical clearance was calculated as follows

$$\text{DPPH inhibition(\%)} = 1 - \frac{A_2 - A_1}{A_0} \times 100\%$$

whereas  $A_0$ ,  $A_1$ , and  $A_2$  were the absorbance of ethanol mixed with DPPH, sample mixed with ethanol, and sample mixed with DPPH, respectively.

**2.8. Hemolysis Experiment.** After washing with PBS and dilution, 1 mL of red blood cells was added to a preheated centrifuge tube with 200  $\mu$ L of hydrogel at the bottom and incubated at 37 °C for 1 h. The centrifuge tubes were then centrifuged at 2000 rpm for 5 min, and the supernatant was transferred to a 96-well plate for the absorbance measurement at 545 nm by a microplate reader. Deionized water and PBS were used as positive and negative controls, respectively.

**2.9. Cytotoxicity Evaluation.** The hydrogel was irradiated by UV light for 2 h and then used to prepare the hydrogel extract according to the previous report.<sup>30</sup> HUVEC (human umbilical vein endothelial

cells, China Center for Type Culture Collection, Wuhan, China) and L929 (a mouse fibroblast cell line, Chinese Academy of Sciences Cell Bank, Shanghai, China) cells were cultured in DMEM containing 10% bovine serum and 1% penicillin/streptomycin. Cells were cultured in a 96-well plate at a density of  $1 \times 10^4$ /well for 24 h, and the medium was then replaced by a new 100  $\mu$ L medium containing a hydrogel extract and continued to incubate for 24 and 48 h. 10  $\mu$ L of CCK-8 solution was added to the well plate and continued to be incubated for 1 h. Subsequently, the microplate reader was used to detect the optical density of each well at 450 nm.

The biocompatibility of the hydrogel was also observed by live/dead staining.  $1 \times 10^5$ /well HUVEC and L929 cells were inoculated in 12-well plates, cultured to adhere to the wall, and then incubated with the hydrogel extract for 24 and 48 h. The Calcein-AM/PI double stain kit was used for live/dead fluorescence staining of cells. The cells inside the pore plate were washed, stained, and observed under an inverted fluorescence microscope according to the instructions.

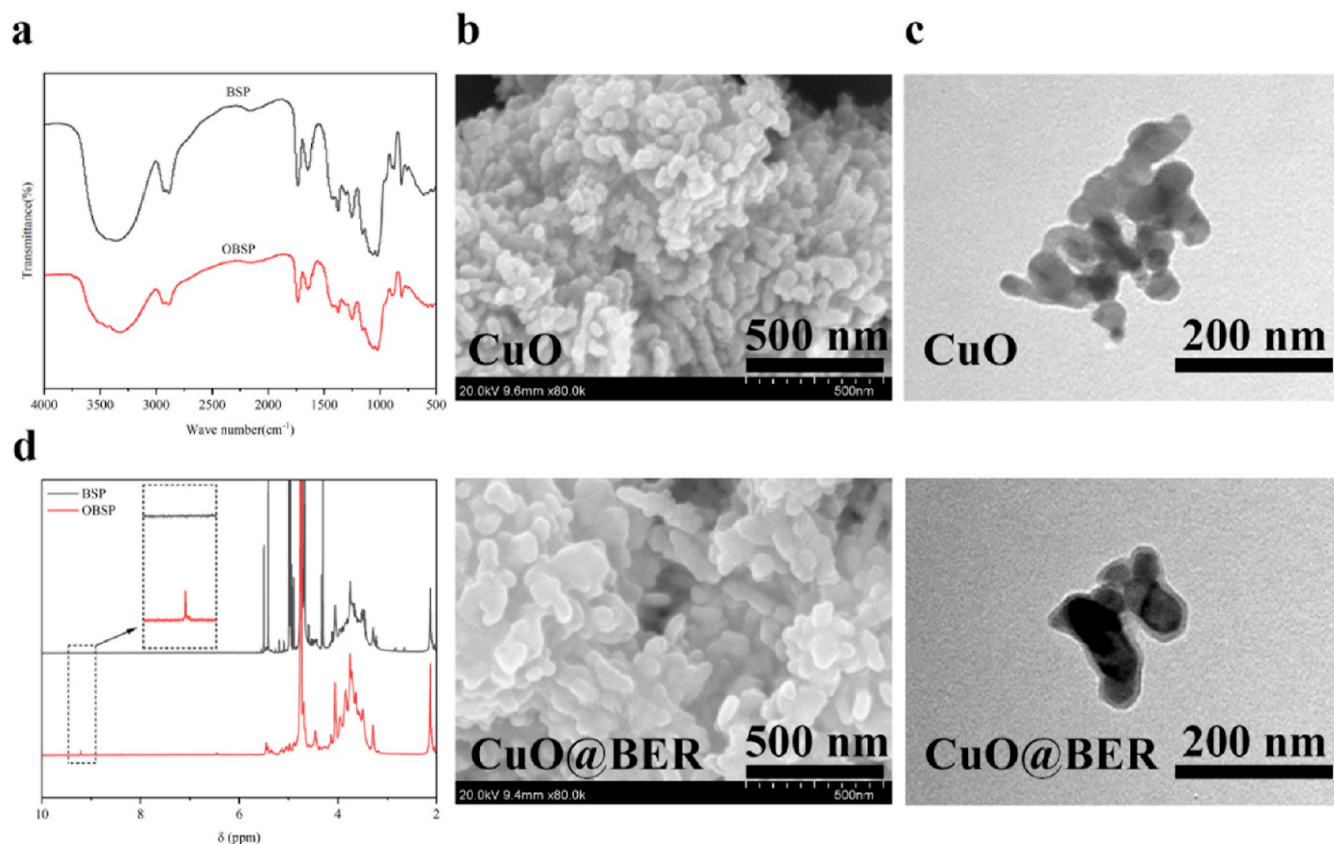
**2.10. Cell Migration Activity.** Scratch test was used to evaluate the effect of hydrogel on cell migration.  $1 \times 10^5$ /well HUVEC and L929 cells were inoculated and cultured in the 12-well plate for 24 h. The medium in the well plate containing the cell scratch area caused by a 200  $\mu$ L pipet tip was replaced by new culture medium containing hydrogel extract. The scratch area was recorded at 0 and 24 h.

**2.11. In Vitro Anti-inflammatory and Antioxidant Activities.** RAW264.7 cells (Chinese Academy of Sciences Cell Bank, Shanghai, China) were inoculated at a density of  $1 \times 10^5$ /well and cultured in a six-well plate for 24 h. After the medium was replaced with the hydrogel extract and incubated for 2 h, 1  $\mu$ g/mL LPS was added and cultured for 24 h. The culture medium was collected, and concentrations of TNF- $\alpha$ , MCP-1, and IL-6 were measured using FAC scanning flow cytometry. Following the same treatment to RAW264.7 cells, the intracellular ROS levels were measured according to the reagent instructions.

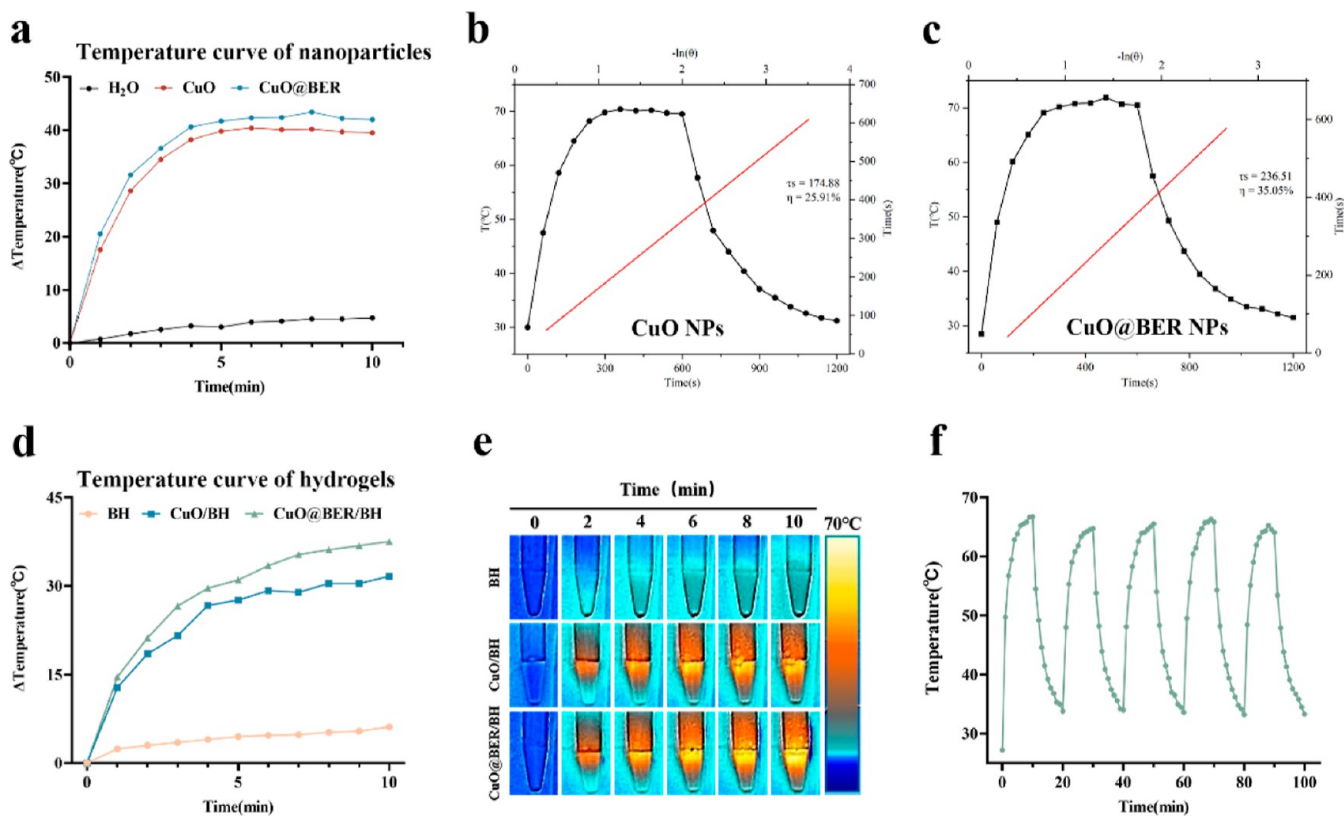
**2.12. Animal Experiment.** To assess the *in vivo* biocompatibility of the hydrogel, the BH hydrogel and CuO@BER/BH hydrogel (100  $\mu$ L) were subcutaneously injected into the backs of ICR mice. On the 14th day post-implantation, the mice that received the hydrogel were euthanized, and skin tissue surrounding the injection site was collected for inflammation analysis using hematoxylin and eosin (HE) staining. Furthermore, organs such as the heart, liver, spleen, lungs, and kidneys were harvested on the 14th day for HE staining. Concurrently, the activities of ALT, AST, and ALP in blood samples collected from the mouse hearts were measured to assess liver toxicity.<sup>31</sup>

ICR mice (6–8 weeks, 18–22 g, Zhejiang Chinese Medical University Animal Experimental Research Center) were used for the animal wound healing experiment. All animal studies were conducted in accordance with the regulations and guidelines of the Zhejiang Chinese Medical University's Institutional Animal Care and Use Committee (IACUC, 20230515-22) and were approved by the IACUC. STZ (10 mg/mL, 150 mg/kg) solution was injected into the peritoneum after fasting to establish type 1 diabetes mice. Fasting blood sugar level that exceeded 16.7 mmol/L after 5 days of STZ administration was considered as the establishment of diabetes models.<sup>32</sup> After the mice were anesthetized by intramuscular injection of 0.04 mg/kg Atropine and intraperitoneal injection of 80 mg/kg Zoletil 50, the hair on the back was removed, and a round wound with a diameter of 1 cm was created with the surgical scissors. 100  $\mu$ L of *S. aureus* bacterial solution ( $1 \times 10^7$  CFU/mL) was added to the wound site for 2 consecutive days to obtain infected diabetes wounds. The mice were randomly divided into four groups (negative group, ANSON Ag Gel group, BH group, and CuO@BER/BH group) and treated with the corresponding hydrogels and NIR radiation (1 W/cm<sup>2</sup>, 5 min). The hydrogels were replaced every 3 days. The wound area was photographed and recorded on 1, 4, 7, 10, and 14 days, and the data were processed by ImageJ software.

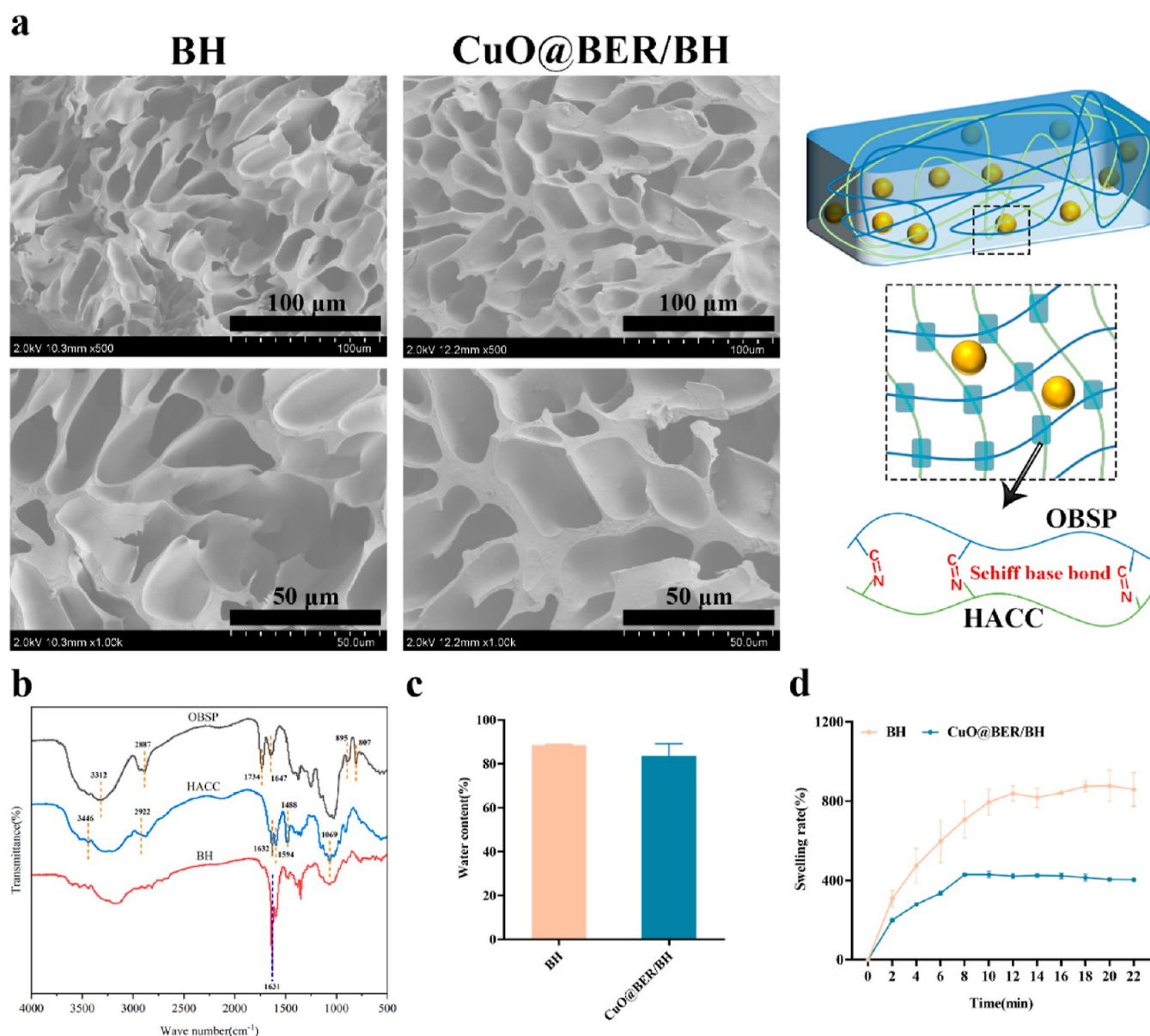
After the last recording of the wound, the mice were euthanized (100 mg/kg, pentobarbital sodium), and skin tissues and major organs, such as heart, liver, spleen, lungs, and kidneys, were collected. The skin tissues were stained with HE and Masson trichrome. Optical



**Figure 1.** FTIR (a) and  $^1\text{H}$  NMR (d) spectra of BSP and OBSP; SEM (b) and TEM (c) images of CuO NPs and CuO@BER NPs.



**Figure 2.** Photothermal properties of the NPs and hydrogels. The heating curve of  $\text{H}_2\text{O}$  and 1 mg/mL NPs (a) and the heating–cooling curve and photographic conversion efficiency of CuO NPs (b) and CuO@BER NPs (c); heating curve (d) and thermal image (e) of hydrogel; temperature curve of CuO@BER/BH hydrogel under NIR on–off cycle (f). The intensity and wavelength of the NIR were  $1 \text{ W}/\text{cm}^2$  and 808 nm.



**Figure 3.** SEM images of hydrogels at different scales (a); FTIR spectra of hydrogels and raw materials (b); and water content (c) and swelling rate (d) of hydrogels.

microscopy was used to observe and record the proliferation of collagen fibers formed by local neovascularization and granulation tissue in wounds. The main organs were stained with HE to investigate the safety of the hydrogel.

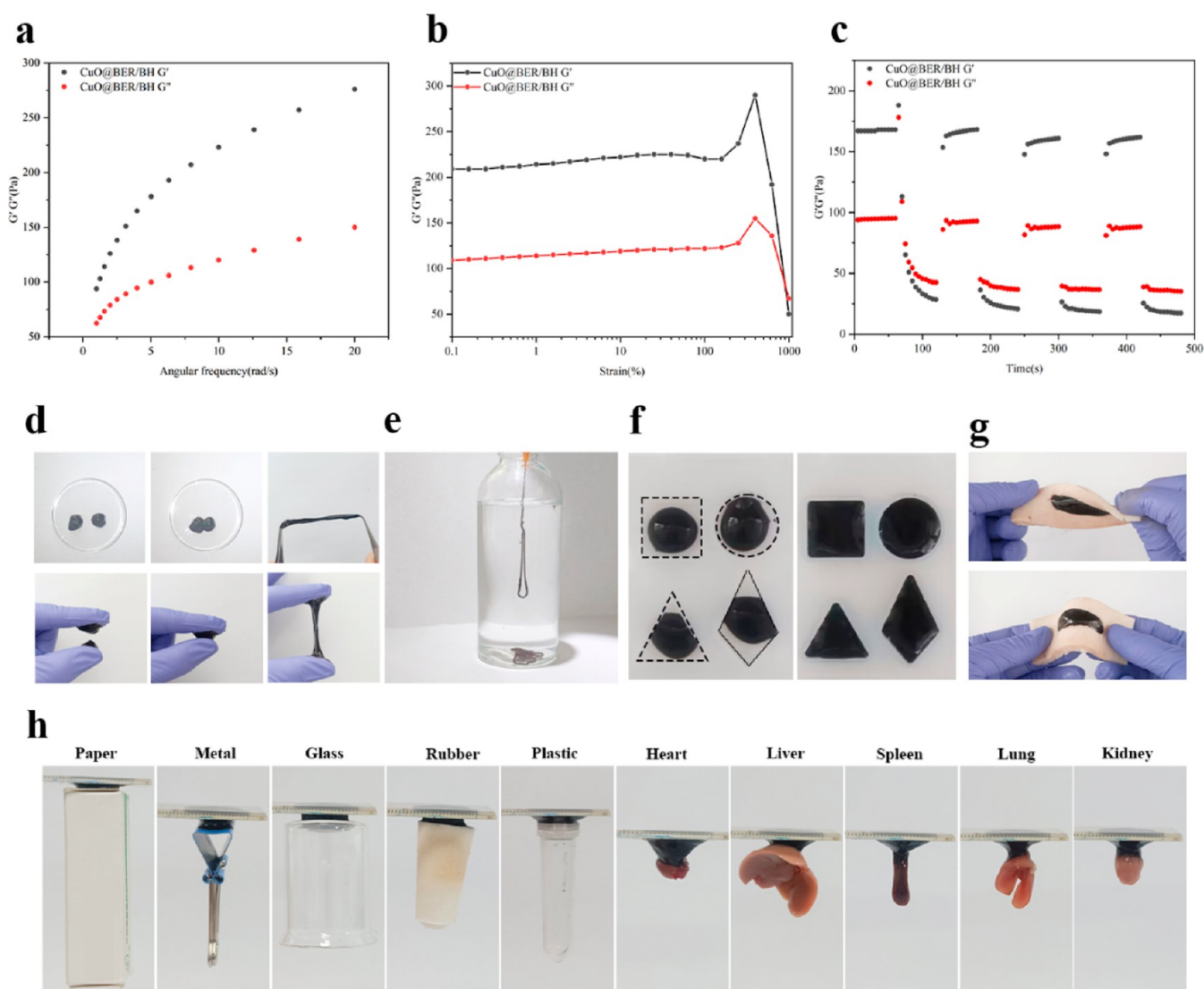
**2.13. Statistical Analysis.** Statistical analysis using SPSS 20.0 was performed on the experimental results, which were expressed as mean  $\pm$  standard deviation (SD), and one-way ANOVA with probabilities of  $*p < 0.05$  and  $**p < 0.01$  displayed the level of statistical significance.

### 3. RESULTS AND DISCUSSION

**3.1. Preparation and Characterization of OBSP and CuO@BER NPs.** The multifunctional composite hydrogel was prepared by a two-step method, including the preparation of dynamic cross-linked hydrogel matrix and loading of photo-thermal particles. The dynamic covalent imine bond can be obtained by the spontaneous cross-linking of amino and aldehyde groups at room temperature.<sup>33</sup> To obtain dialdehyde groups,  $\text{NaIO}_4$  was used to oxidize o-diol on the sugar unit of

BSP, and FTIR and  $^1\text{H}$  NMR were used to characterize BSP and OBSP. The results are illustrated in Figure 1. In the FTIR spectrum (Figure 1a), the absorption peaks at 3355 and 2887  $\text{cm}^{-1}$  were attributed to the vibrations of  $-\text{OH}$  and  $\text{C}-\text{H}$ , the tensile vibrations of  $\text{C}=\text{O}$  occurred at 1732 and 1647  $\text{cm}^{-1}$ , and the absorption peaks of mannose and  $\beta$ -glucose residues appeared at 810 and 875  $\text{cm}^{-1}$ , respectively.<sup>34</sup> In the  $^1\text{H}$  NMR spectrum (Figure 1d), the proton signal belonging to the aldehyde group appeared at 9.22 ppm of OBSP, indicating the successful modification of BSP. Hydroxylamine hydrochloride potentiometric titration was used to determine the oxidation degree of OBSP, and the results showed that the oxidation degree was  $16.2 \pm 0.81\%$ .

Dopamine coating and a  $\pi-\pi$  conjugation system were used to prepare CuO@BER nanoparticles. The polydopamine formed by the self-polymerization of dopamine was rich in catechol structure and can be attached to the surface of CuO nanoparticles. At the same time, BER was immobilized on the



**Figure 4.** Rheological properties of CuO@BER/BH hydrogel in angular frequency scanning (a), strain scanning (b), and multiple strain scanning cycles (c); self-healing (d), injectability (e), shape adaptability (f), close fitting with pigskin (g), and adhesion to the surfaces of different objectives (h) of CuO@BER/BH hydrogel.

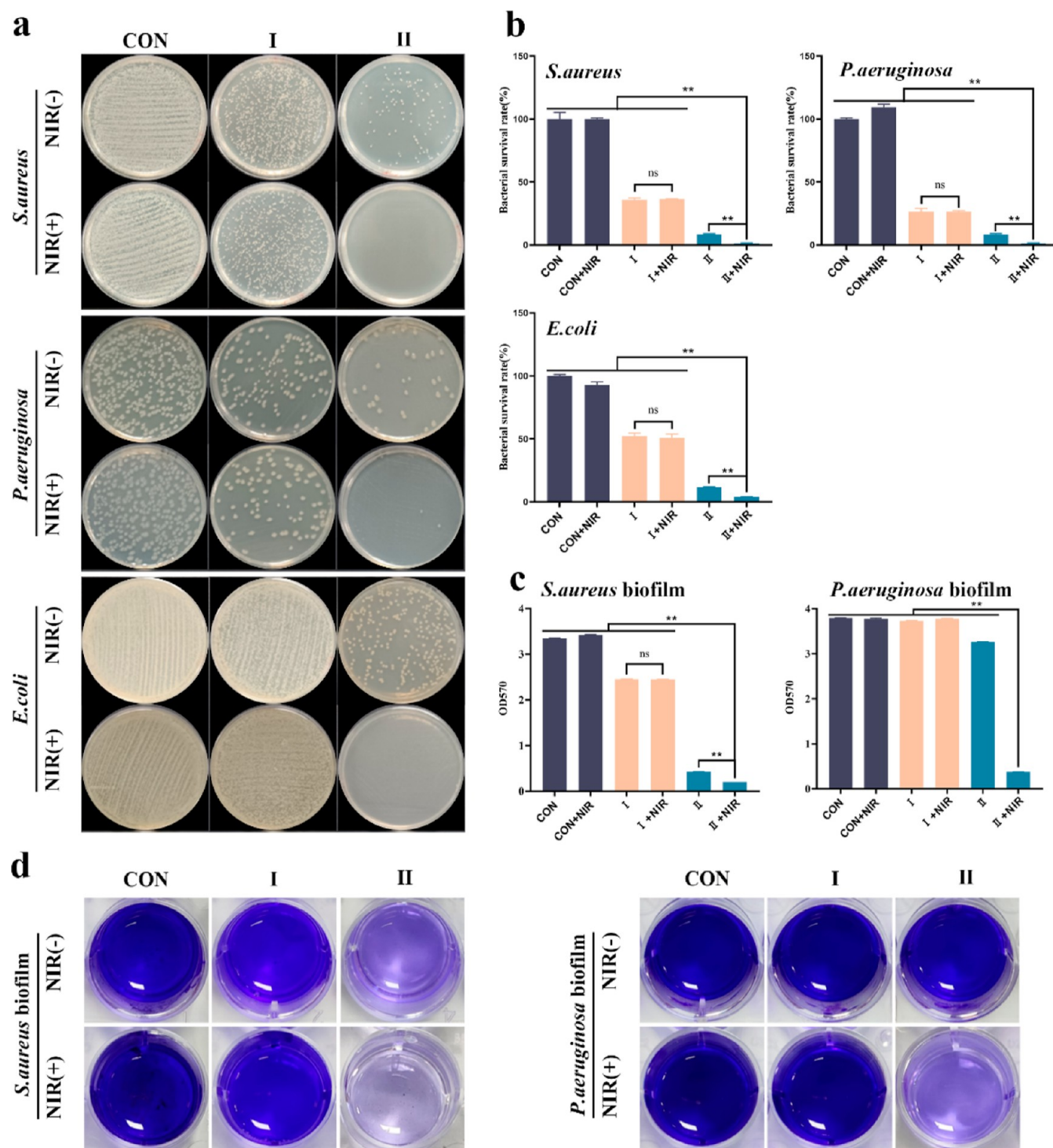
surface of nanoparticles by  $\pi$ - $\pi$  stacking with the aromatic structures of BER.<sup>35</sup> In the SEM and TEM images (Figure 1b,c), CuO@BER particles exhibited a slightly larger diameter than CuO, with a different transmittance at inner and outer layers, and CuO showed uniform transmittance, indicating that there was an obvious core-shell structure in CuO@BER (Figure 1c). In the EDS results (Figure S1), CuO@BER showed the signal peaks of Cu, O, N, and C. The above results indicated that dopamine and BER were successfully loaded on CuO nanoparticles.

**3.2. Near-Infrared Photothermal Characteristics.** The photothermal properties of CuO and CuO@BER NPs are illustrated in Figure 2a. The temperature change ( $\Delta T$ ) of water was not significant, while the temperature of NPs increased rapidly within 4 min, and this trend gradually slowed down until reaching the plateau period approximately at 5–6 min. Meanwhile, CuO@BER had a higher heating effect than CuO NPs. Further calculation of  $\eta$  showed that the photothermal conversion efficiency of CuO and CuO@BER NPs were 25.91 and 35.05%, respectively (Figure 2b,c), which were slightly higher than previous reports,<sup>36–38</sup> indicating the enhanced

photothermal characteristics established in this study by the integration of dopamine and BER to CuO.

In addition to the NPs, the photothermal effect of the composite hydrogels was also investigated (Figure 2d,e). With the advance of NIR irradiation time, the temperature of BH hydrogel rose slightly, while the temperature of the hydrogel loaded with CuO and CuO@BER NPs changed significantly. The temperature of CuO/BH and CuO@BER/BH rose rapidly within 4 min, and the subsequent rise of temperature gradually slowed down. Throughout the entire period, the temperature rise of CuO@BER/BH was more significant than that of CuO/BH, which was consistent with our previous results of NPs. The photothermal stability of CuO@BER/BH was investigated using continuous NIR irradiation and five cycles of closure, and the result (Figure 2f) showed a repeatable temperature curve during the intermittent irradiation to CuO@BER/BH, which proved the photothermal stability of CuO@BER/BH.

**3.3. Preparation and Characterization of Hydrogels.** After OBSP and HACC were mixed, the hydrogel was obtained by the Schiff base reaction between the aldehyde



**Figure 5.** Representative colony formation of bacteria on plates treated with NS and different hydrogels with or without NIR irradiation for 5 min, respectively (a); quantitative results of bacterial survival rate obtained by bacterial liquid turbidity method (b); quantitative detection results of *S. aureus* and *P. aeruginosa* biofilm (c); image of bacterial biofilm stained with crystal violet (d); I and II represent BH and CuO@BER/BH, \* represents  $p < 0.05$ , \*\* represents  $p < 0.01$ , and ns represents no statistically significant difference.

groups on OBSP and amino groups on HACC. The nanospheres were uniformly fixed in the 3D network of the hydrogel. SEM was used to observe the microstructure of the hydrogels. As shown in Figure 3a, each hydrogel had a well-proportioned three-dimensional porous structure, and a small number of rough particles were observed in the porous structure of CuO@BER/BH, attributing to the CuO@BER NPs. In the FTIR spectra of OBSP, HACC, and hydrogels

(Figure 3b),  $3446\text{ cm}^{-1}$  of HACC molecule was assigned to –OH and N–H symmetric stretching vibration and intra-molecular hydrogen bond,  $2922\text{ cm}^{-1}$  corresponded to C–H stretching vibration,  $1632$  and  $1594\text{ cm}^{-1}$  corresponded to C=O (amide I) and N–H (amide II) stretching vibrations,  $1488\text{ cm}^{-1}$  belonged to the unique characteristic peak of quaternary ammonium salt methyl bond, and  $1069\text{ cm}^{-1}$  was attributed to C–O stretching vibration. The C=O bond at

1732  $\text{cm}^{-1}$  in OBSP disappeared in the hydrogel, which may be related to the reaction between the aldehyde group and amino group. At the same time, the stretching vibration of the  $\text{C}=\text{N}$  double bond observed at 1631  $\text{cm}^{-1}$  in the prepared hydrogel demonstrated the occurrence of Schiff base reaction between OBSP and HACC.

**3.4. Water Content and Swelling Properties of Composite Hydrogels.** High water content is an important feature of hydrogels,<sup>39</sup> and both hydrogels contained more than 80% of water, as shown in the results in Figure 3c. The swelling performance of the hydrogels is shown in Figure 3d. The hydrogels expanded rapidly in the first 2 min, and then the water absorption and swelling rate slowed down, reaching a plateau after 8 or 10 min. The final results of the swelling rate of CuO@BER/BH were higher than 400%, indicating that the hydrogel had excellent water absorption ability and can rapidly absorb the exudate of the wound, which although is not the primary issue in most of the diabetic wounds.<sup>28</sup> The experimental results revealed a higher swelling rate of the BH hydrogel, which was possibly attributed to the detachment of the nanoparticles from the CuO@BER/BH hydrogel during the measurement, causing a significant weight loss and a lower calculated swelling rate in comparison to the BH hydrogel. In the stability experiment (Figure S2), the BH and CuO@BER/BH hydrogels had good stability in neutral and weakly alkaline solutions and exhibited the lowest stability in weakly acidic solutions. This phenomenon was likely due to the weak stability of the Schiff base structure of the hydrogels in weakly acidic environments.<sup>40</sup>

**3.5. Rheology Characteristics of Hydrogels.** Viscoelasticity is an important mechanical property of a hydrogel, which reflects the energy storage and dissipation capacity. The hydrogel modulus was studied by angular frequency scanning from 1 to 20 rad/s, and the experimental results showed that the loss modulus ( $G''$ ) of the composite hydrogel increased but not exceeding the storage modulus ( $G'$ ) within the test frequency range (Figure 4a), indicating the successful preparation and stable mechanical properties of the hydrogel.<sup>41</sup> Meanwhile, there was no intersection of the  $G'$  and  $G''$  curves, indicating that the composite hydrogel did not transform into a liquid state in the measured angular frequency range and was mechanically stable. The results of strain scanning (Figure 4b) showed a gentle rise of  $G'$  and  $G''$  of the composite hydrogel in the strain range from 0.1 to 100% and a sharp increase to the maximum around the strain of 400%, after which  $G'$  and  $G''$  started to decline and intersected to  $G'' > G'$  at the strain of 900%, indicating the transition of the hydrogel to the liquid-state response.<sup>42</sup>

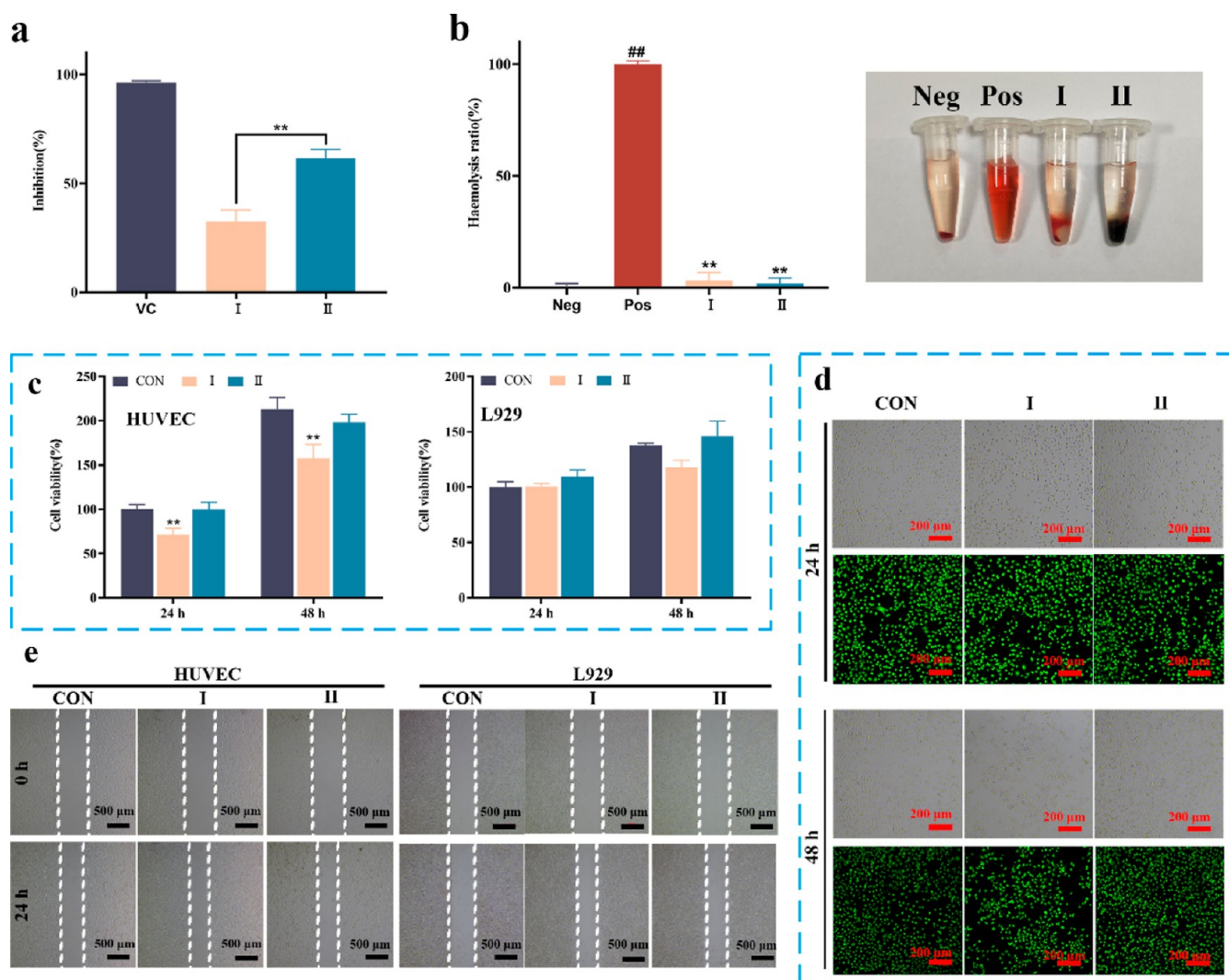
The self-healing properties of the composite hydrogel were evaluated by a cyclic strain test. As shown in Figure 4c, the hydrogel presented  $G' > G''$  at the applied strain of 1% in the first 60 s. The dramatic change of strain from 1 to 1000% during 60 to 120 s caused the sharp drop of the  $G'$  and  $G''$  values and resulted in  $G'' > G'$ , indicating that the hydrogel network was damaged. The values of  $G'$  and  $G''$  turned back to the initial value after the applied strain returned to 1%, implying that the network cross-link of the hydrogel had been successfully restored. The process of restoration and destruction of the hydrogel networks was successfully repeated at least four times, indicating the self-healing ability of the hydrogel, and the reversible Schiff base bond between OBSP and HACC was considered to be the contributor for this rheological property.<sup>19,33</sup>

**3.6. Macroscopic Self-Healing, Injectability, Shape Adaptation, and Adhesion Properties of Hydrogels.** As illustrated in Figure 4d,e, two separated hydrogels fused into one after splicing without additional stimulation, and the formed hydrogel withstood certain tension without breaking at room temperature. Meanwhile, the composite hydrogel could be extruded from the syringe under the action of an external force. These experiments successfully demonstrated the self-healing and injectability properties of the composite hydrogels and confirmed the hydrogels' ability to withstand external tension and mechanical stress, thereby preserving the integrity of the protective barrier between the wound and the environment. This integrity is crucial in preventing wound infections that may arise from external bacterial invasion.<sup>43</sup>

As depicted in Figure 4f, upon addition to grooves of varying shapes, the hydrogel gradually filled the mold at room temperature, which indicated its shape adaptability. At the same time, the composite hydrogel adhered to the pigskin could produce corresponding deformation with the twisting and bending of the pigskin and maintain the original fitting shape (Figure 4g). The composite hydrogel was adhesive not only to the skin but also to other biological (animal organs) or nonbiogenic (paper, iron, glass, rubber, and plastic) substances, as shown in Figure 4h. Shape adaptability and adhesive ability of the hydrogel facilitate its application to wounds with irregular shapes, which might not only provide physical isolation and wet environment for defects but also provide the substrate continuous and effective drug release to further enhance the potential of hydrogels as wound dressings.<sup>17</sup> Integrating the self-healing, injectability, shape adaptability, and adhesiveness characteristics of the hydrogel might help to protect the wound from drug injury during exercise and dynamic wound care.<sup>44</sup>

**3.7. Antibacterial Activity and Biofilm Inhibitory Activity of Hydrogels.** The antibacterial results of hydrogel by the spread plate method are shown in Figure 5a. Compared with the control group using sterile physiological saline, BH showed significant antibacterial effects on *S. aureus*, *P. aeruginosa*, and *E. coli*, where a relatively strong effect was found on *S. aureus* and a relatively weak effect was found on *E. coli*. The addition of CuO@BER NPs further enhanced the antibacterial effects against three types of bacteria. Further NIR treatment results showed that individual NIR irradiation had almost no antibacterial effect on bacteria and did not affect the antibacterial effect of BH. In contrast, CuO@BER/BH showed significant inhibitory effects on three types of bacteria under NIR treatment. The morphology of bacteria during the above treatment was investigated using SEM. The results in Figure S4a showed that the bacteria of sterile physiological saline control group maintained a complete membrane structure, while severely distorted and fragmented bacteria were found in the CuO@BER/BH + NIR group. The antibacterial effect of the hydrogel on bacteria was further evaluated with live/dead fluorescence staining. Red fluorescence was hardly found in the sterile physiological saline control group but obviously presented in the CuO@BER/BH + NIR group (Figure S4b).

The turbidity of the bacterial solution was also used to evaluate the antibacterial properties of hydrogels. Compared with the control group, all hydrogels presented an inhibitory effect on the bacterial growth (bacterial survival rate). Among the different groups without NIR treatment, CuO@BER/BH showed a much stronger inhibitory effect on three kinds of bacteria in comparison with the BH group. NIR treatment



**Figure 6.** Image of DPPH clearance results of different hydrogels (a); quantitative results of hemolysis rate (b); and cell viability of HUVECs and L929 cells incubated with different hydrogel leachates (c). Live/dead staining images of HUVEC cells incubated with different hydrogel leachates for 24 and 48 h (live cells: green, dead cells: red) (d); cell scratch results of HUVEC and L929 cells incubated with different hydrogel leachates for 24 h (e); I and II represent BH and CuO@BER/BH, compared to Neg, ## represents  $p < 0.01$ , and compared to Pos, \*\* represents  $p < 0.01$ .

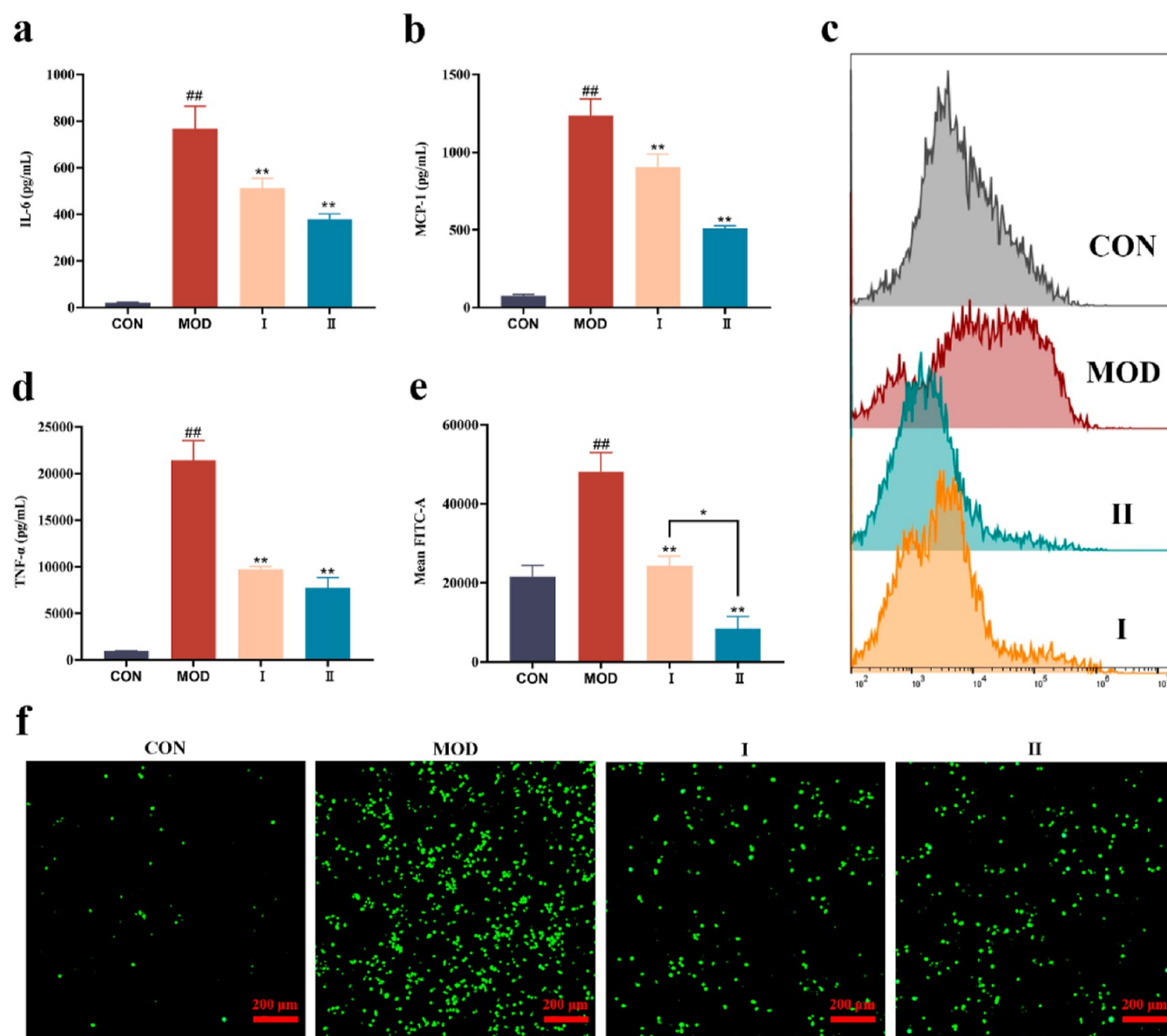
significantly enhanced the antibacterial effect of CuO@BER/BH but showed no effect on the BH group. These results indicated that the photothermal effect of CuO@BER NPs irradiated by NIR could further improve the antibacterial performance of the hydrogel. The hydrogel without NPs showed antibacterial effects against *S. aureus* and *P. aeruginosa* because the quaternary ammonium group of HACC could adsorb bacteria via electrostatic action and destroy the inherent integrity of the bacterial cell wall and membrane.<sup>45</sup> The improvement of the antibacterial effect of NIR irradiation on CuO@BER/BH was mainly attributed to the action mechanism of PTT such as heat, nano-perforation, membrane stability destruction, biomolecular binding, and oxidative damage.<sup>46</sup>

Biofilm is a complex 3D structure of grouped bacteria adhering to the surface and embedding in the automatically generated matrix of extracellular polymer materials, and it can further hinder the healing of chronic wounds by enhancing the resistance of bacteria to antibiotics to a floating state 1000 times.<sup>47,48</sup> The delayed wound healing caused by biofilms has been widely reported, posing a great challenge to the

treatment.<sup>49–51</sup> To investigate the antibiofilm effect of the PTT hydrogel developed in our study, crystal violet staining experiments were conducted. As illustrated in Figure 5d, deep bluish violet was observed in the blank group, BH group, and CuO@BER/BH groups, but almost not in the CuO@BER/BH groups irradiated by NIR, indicating that the PTT treatment had a good inhibitory effect on the biofilm. The low absorbance of the CuO@BER/BH + NIR group at 570 nm (Figure 5c) also confirmed the significant inhibitory effects of PTT treatment on the biofilm. The satisfactory activity of the antibiofilm provided an effective solution to the infection in chronic wounds.<sup>46</sup>

### 3.8. DPPH Radical Scavenging Activity of Hydrogels.

The results of the DPPH scavenging capacity of the hydrogels are shown in Figure 6a. The relative DPPH scavenging rates of BH and CuO@BER/BH were  $32.46 \pm 5.24\%$  and  $61.66 \pm 3.94\%$ , respectively. The inhibition of CuO@BER/BH was significantly higher than that of BH hydrogel, which indicated that the addition of CuO@BER enhanced the DPPH scavenging capacity of BH. DPPH free-radical scavenging ability correlated to the antioxidant activity,<sup>52,53</sup> and oxidative



**Figure 7.** Effects of different hydrogels on the contents of IL-6 (a), MCP-1 (b), and TNF- $\alpha$  (d) in RAW264.7 cells stimulated by LPS; flow analysis (c) and quantitative results (e) of ROS removal performance of different hydrogels; ROS fluorescence detection images by DCFH-DA processing (f); I and II represent BH and CuO@BER/BH. Compared to CON, ## represents  $p < 0.01$ , and compared to MOD, \*\* represents  $p < 0.01$ .

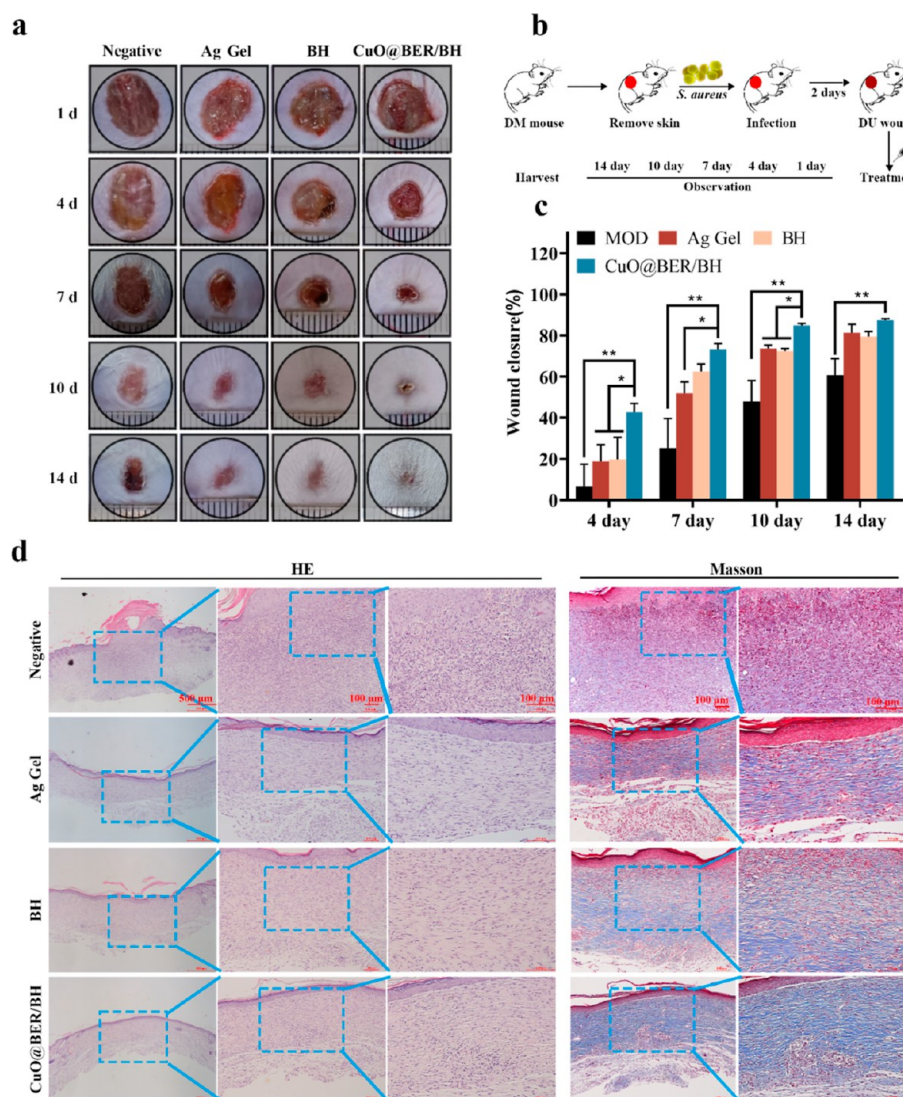
stress promoted the chronic wound site to delay healing by affecting the cell migration, angiogenesis, and the phase transition from inflammation to proliferation,<sup>54</sup> so the prepared composite hydrogel with antioxidant properties might be beneficial to wound healing.<sup>55,56</sup>

**3.9. Evaluation of the Effect of Hydrogel on Cell Activity.** **3.9.1. Cytocompatibility Evaluation of Hydrogels.** The dressing would inevitably contact the blood at the wound site during application; therefore, the hemolytic activity of the hydrogel was measured. The results (Figure 6b) showed that there was no erythrocyte precipitation after centrifugation, and the supernatant was red in the group treated with deionized water. The erythrocyte precipitation and relatively clear supernatant were observed in the PBS and hydrogel treatment groups, indicating nonhemolysis. At the same time, the further test results of the supernatant showed that the hemolysis rates of all hydrogel groups were lower than 5%, indicating the good blood compatibility of the prepared hydrogel.<sup>57</sup>

L929 cells and HUVEC cells were incubated with extracts from hydrogels for 24 and 48 h, and the cell viability remained above 70%, as illustrated in Figure 6c, representing nonobvious cytotoxicity.<sup>58</sup> From the live/dead cell staining (Figures 6d and S5) experiments, it could be observed that the cell morphology of the hydrogel treatment group did not change significantly compared with the normal control group in the bright field. No obvious difference of green or red fluorescence between the control group and hydrogel-administered groups was observed.

The results of the above experiments were consistent with the quantitative results of cell viability and low hemolysis rate of hydrogels, all of which confirmed the satisfactory biocompatibility and biological safety of the hydrogel prepared in wound application.

**3.9.2. Cell Migration Activity.** The ability of cell migration plays a crucial role in wound healing, so ideal skin repair materials need to have a certain role in promoting cell migration.<sup>59</sup> As the results show in Figure 6e, after 24 h



**Figure 8.** Representative images of infected wounds on days 1, 4, 7, 10, and 14 (a); schematic illustration of establishing a model of *S. aureus* infected mice and the process of different hydrogel treatments (b); quantitative results of wound closure area for all groups on days 4, 7, 10, and 14 (c); HE staining and Masson's trichrome staining images of skin tissue treated with different dressings on day 14 (d); \* represents  $p < 0.05$ , and \*\* represents  $p < 0.01$ .

treatment with culture medium or extract, cells of each group migrated to the scratch area. Compared with the blank control group, the cell migration effect of the hydrogel treatment group was slightly strengthened, indicating that the hydrogel had the possibility of promoting cell migration and had potential in the application of wound healing.

**3.9.3. Anti-inflammatory Activity.** The inhibitory effect of hydrogel on the inflammation of RAW264.7 cell induced by LPS is exhibited in Figure 7a,b,d. After LPS stimulation, the level of proinflammatory cytokines increased significantly compared with the normal control group, which indicated successful modeling. Compared with the model group, the administration of hydrogel extracts significantly reduced the level of proinflammatory factors, indicating the inhibitory effect of the hydrogel on cellular inflammation. The anti-inflammatory activity could be due to the pharmacological effect of BSP and BER in the hydrogel. Long et al. found that BSP inhibited the activation of proinflammatory cytokines IL-6 and TNF- $\alpha$ , and these effects followed a dose-dependent manner.<sup>60</sup> Chen et al. reported that BER exhibited an inhibitory effect on LPS-

induced inflammatory response by regulating the NF- $\kappa$ B/Nlrp3 signaling pathway in the acute lung injury model.<sup>61</sup> Excessive and prolonged inflammation is considered as the critical cause of delayed healing and scar formation, and the hydrogel prepared in this study could effectively reduce the level of proinflammatory cytokines, which was beneficial for alleviating the excessive and overlonged inflammation in chronic wound that hinders the healing of the wound.<sup>62</sup>

**3.9.4. Antioxidant Activity.** Persistent hyperglycemic environment promotes the production of ROS, thereby promoting the inflammation cascade reaction and stimulating oxidative stress, thus forming a vicious circle and eventually leading to chronic wounds.<sup>13</sup> Inspired by the antioxidant activity of hydrogels in free-radical scavenging experiments, we further measured their intracellular antioxidant activity using the DCFH-DA probe. As shown in Figure 7f, the quantity and intensity of fluorescence dots significantly increased after LPS stimulation; however, the increment was reduced by the treatment of hydrogel extracts. The quantitative results of fluorescence intensity measured by flow cytometry illustrated

in Figure 7e showed that the hydrogel extract treatment significantly reduced the elevated value caused by LPS stimulation and confirmed the above microscopic observations. Additionally, the value of FITC-A in the CuO@BER/BH group was significantly lower than that of the BH group, which could be attributed to the antioxidant activity of BER.<sup>14</sup> It has been reported that the control of ROS level could alleviate the oxidative damage at the wound site, promote the proliferation of epithelial cells, angiogenesis, and tissue restoration; therefore, the antioxidant activity of CuO@BER/BH hydrogel was beneficial to the wound healing.<sup>63</sup>

The increase in the number of M1 macrophages and neutrophils in chronic wound sites is closely related to the elevated levels of ROS, chemokines, and cytokines, and these factors mutually promote the inflammation, forming a constant vicious cycle<sup>64</sup> and hindering wound healing. Combined with the abilities to regulate the microenvironment of the wound, including the regulation of the inflammatory cytokine levels, reduction of ROS, and promotion of cell migration, the hydrogel was expected to possess good capability that promotes the healing of chronic wounds.

**3.10. In Vivo Wound Healing Performance.** The results of subcutaneous injection of hydrogel into the back of mice for 14 days are shown in Figure S6, where section a displays the HE staining results of animal organ tissues, section b shows the serum results of ALT, AST, and ALP, and section c shows the HE staining results of the skin tissue around the injection site. The results indicated that, compared with the normal control group, the tissue sections of organs in the hydrogel-treated group did not exhibit significant changes, and the levels of ALT, AST, and ALP in the serum showed no significant alterations, indicating that the hydrogel was nontoxic to mice and demonstrated good biocompatibility.

The DU model with *S. aureus* infection was established on ICR mice to evaluate the healing promoting effect of the hydrogels. As shown in Figure 8a, in the first 10 days, compared with other groups, the wound area in the CuO@BER/BH treatment group decreased the most significantly; however, on the following 14th day, the wound area of the CuO@BER/BH group did not change significantly, which may be because the healing stage entered the subsequent proliferation and remodeling stages. The quantitative result (Figure 8c) showed that the healing rate of the CuO@BER/BH treatment group was close to 85% on the 10th day, which was significantly higher than that of the commercial Ag gel and BH groups, indicating the excellent healing promotion effect of the prepared CuO@BER/BH photothermal hydrogel on the infected wounds of diabetic mice.

The histopathological analysis of the wound skin tissue was further conducted after 14 days, and results are shown in Figure 8d. The control group retained a small area of scab, while the hydrogel treatment group formed a complete, dense, and thin epidermal layer. Meanwhile, the control group showed a significant infiltration of inflammatory cells, with a dense and compact structure and disorderly arrangement of collagen. The situation was significantly improved in the hydrogel treatment groups. Inflammatory cell infiltration could be seen in the commercially available gel (ANSON Ag Gel) and BH, and this was less obvious in the CuO@BER/BH group. CuO@BER/BH group had the most obvious effect, with a small amount of inflammatory cell infiltration in its tissues. The collagen fibers filled the defects in a reticular pattern and were loosely and orderly arranged with the

direction parallel to the skin surface. These results indicated that CuO@BER/BH with PTT could alleviate inflammatory pressure, improve re-epithelialization, collagen deposition, angiogenesis, and promote tissue regeneration. H&E staining was also performed on other major organs (heart, liver, spleen, lung, and kidney), which did not show significant change during the application of hydrogels when compared with the control group, as presented in Figure S7. The results verified that no serious injury to major organs was generated by the developed hydrogels, which were highly nontoxic in vivo.

As shown in Figure S8, during NIR irradiation, the temperature of the BH hydrogel on the mouse body surface did not change significantly, while the temperature of the CuO@BER/BH hydrogel increased rapidly. It has been proved that the tissue cells can withstand a relatively low temperature in the range of 50–60 °C in a short time. At the same time, because the hydrogel has high water content and the ability to maintain water, this temperature will not cause harm to the skin.<sup>65</sup>

The wound healing was a complex process with a series of overlapping stages of hemostasis, inflammation, proliferation, and remodeling, and the overlonged inflammation stages postponed the healing is one of the typical characteristics of DU. After debridement, improper wound care frequently causes infection and inflammation during the clinical treatment of DU. Therefore, functional hydrogels aiming at the above challenges are the demand of the clinical demand. The good healing rate of CuO@BER/BH obtained in animal experiment demonstrated the efficiency of the hydrogel in treating DU, which could be contributed by the excellent antibacterial, antioxidative, and antiinflammation properties verified in in vitro study. Observations from HE-stained tissues confirmed the anti-inflammation properties, preventing a prolonged inflammation stage and promoting the transition of healing to the stage of proliferation and remodeling. Improved migration performance from the in vitro study also explained the accelerated wound healing and skin regeneration observed in the pathological section of animal study.

Some of recent studies reported that advanced glycation end products (AGEs), accumulated in the diabetic patient, were effective oxidants to promote the production of excessive ROS and were considered a risk factor for injury and chronic ulcers.<sup>66</sup> Chen et al. found that BSP inhibited the formation of advanced glycation end products (AGEs) after 15 days of incubation in the BSA glucose system.<sup>67</sup> Therefore, the inhibition of BSP on AGEs may also be one of the mechanisms of the hydrogel to heal diabetic wounds. In addition to above, the disruption of M1 to M2 macrophage transition is a significant factor in the excessive and prolonged inflammation and delayed healing of diabetic wounds.<sup>68</sup> Feng et al. found that the BER effectively promotes the transition from M1 macrophages into M2 macrophages, without altering the phenotype of M0 and M2 macrophages.<sup>69</sup> Therefore, the influence of BER on macrophage polarization might also be one of the potential mechanisms of the excellent healing effect of CuO@BER/BH.

## 4. CONCLUSIONS

In conclusion, an NIR-responsive photothermal hydrogel with remarkable self-healing and injectable properties was developed in this study. The exceptional photothermal characteristics of the hydrogel significantly augmented its antibacterial effectiveness against *S. aureus*, *P. aeruginosa*, and *E. coli*.

Moreover, the incorporation of BSP and BER demonstrated outstanding results in *in vitro* experiments, showcasing excellent cell compatibility, anti-inflammatory, and antioxidant capabilities. These properties were highly promising for potential therapeutic applications. Furthermore, our animal experiments had yielded promising results. The hydrogel had shown its ability to mitigate inflammatory cell infiltration in DU wounds, accelerate re-epithelization, and promote collagen deposition, thus effectively expediting the wound healing process. Considering these significant findings, the prepared hydrogel holds great potential for the advancement of DU dressings. The combination of its photothermal and therapeutic properties makes it a promising candidate for further development and future clinical applications in wound management.

## ■ ASSOCIATED CONTENT

### SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.biomac.4c00013>.

EDS results of CuO NPs and CuO@BER NPs, stability results of hydrogels, rheological properties of BH hydrogel, SEM images and live/dead bacteria staining photographs of CuO@BER/BH, live/dead staining images of L929 cells incubated with hydrogel leachates, results of subcutaneous injection of hydrogels, HE staining results of the main organs of mice in wound healing experiments, and typical thermography of hydrogel during NIR irradiation in animal experiments (PDF)

## ■ AUTHOR INFORMATION

### Corresponding Authors

**Bingqi Zhu** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China; Email: [bingqi\\_zhu@hotmail.com](mailto:bingqi_zhu@hotmail.com)

**Zhishan Ding** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China; [orcid.org/0000-0001-5740-1479](https://orcid.org/0000-0001-5740-1479); Email: [dzsztcm@163.com](mailto:dzsztcm@163.com)

### Authors

**Kai Zhao** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China

**Zhengbo Hu** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China

**Xingcan Chen** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China

**Yuchi Chen** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China

**Mingyuan Zhou** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China

**Xiaoqing Ye** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China

**Fangmei Zhou** – School of Medical Technology and Information Engineering, Zhejiang Chinese Medical University, Hangzhou 310053 Zhejiang, The People's Republic of China

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.biomac.4c00013>

### Author Contributions

B.Z. and Z.D. contributed equally to this paper. K.Z.: conception, design, draft, and revision of the manuscript. Z.H.: data curation. X.C.: data curation. Y.C.: data curation and formal analysis. M.Z.: data curation and formal analysis. X.Y.: data curation and formal analysis. F.Z.: data curation and formal analysis. B.Z. and Z.D.: project administration, supervision, validation, and visualization.

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### Notes

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